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(57) **ABSTRACT**

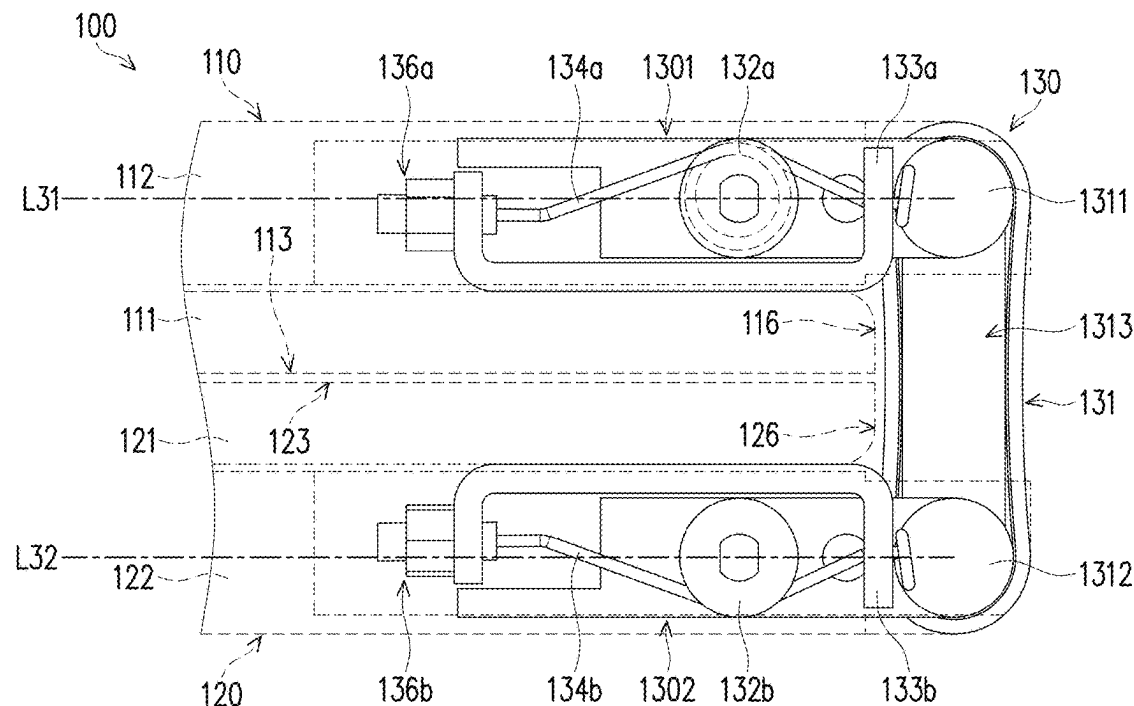
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Related U.S. Application Data

An electronic device including a first body, a second body and at least one pivot module is provided. The pivot module includes a first transmission wheel, a linkage assembly, a first sliding member, a first driven member and a first rope. The linkage assembly is connected to the first body and the second body. The first sliding member is connected to the first body and linked to the first transmission wheel. The first driven member is connected to the first transmission wheel and the linkage assembly. The first rope is connected to the first sliding member and the first transmission wheel. The rotation of the first body drives the first driven member and the linkage assembly to move, thereby driving the first transmission wheel to rotate, so as to drive the first sliding member and at least part of the first body to slide relative to the linkage assembly.

Publication Classification

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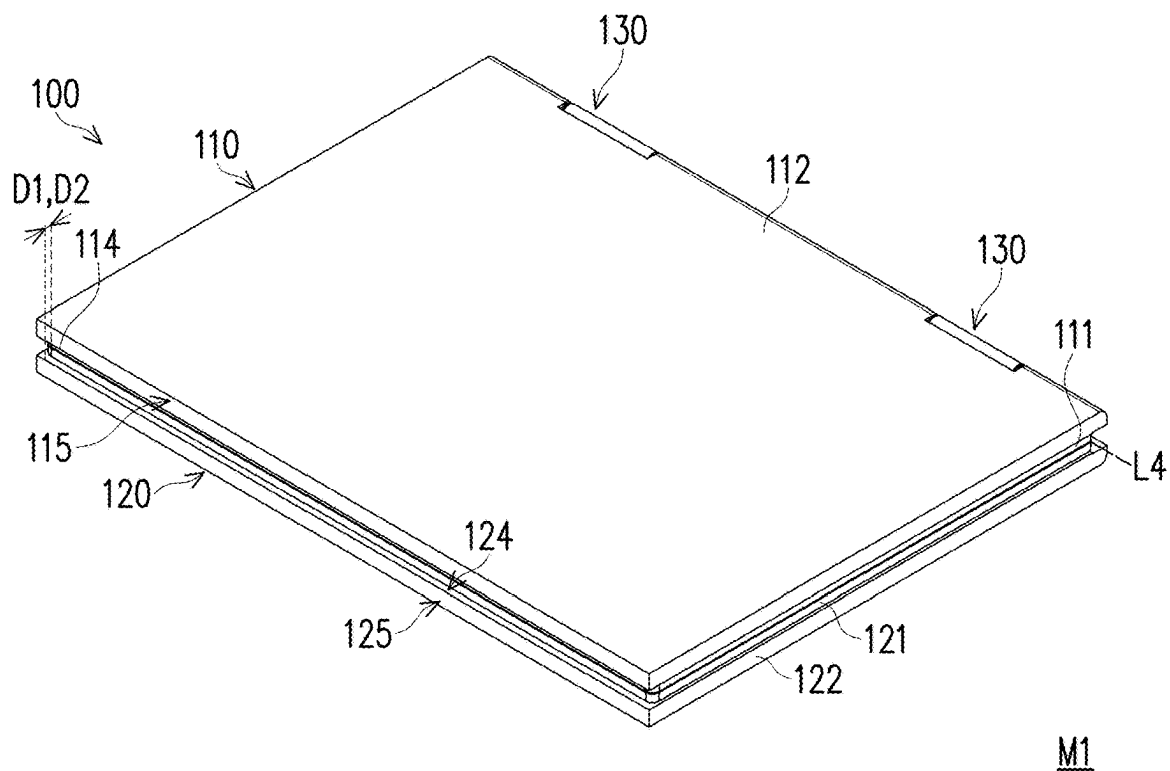


FIG. 1

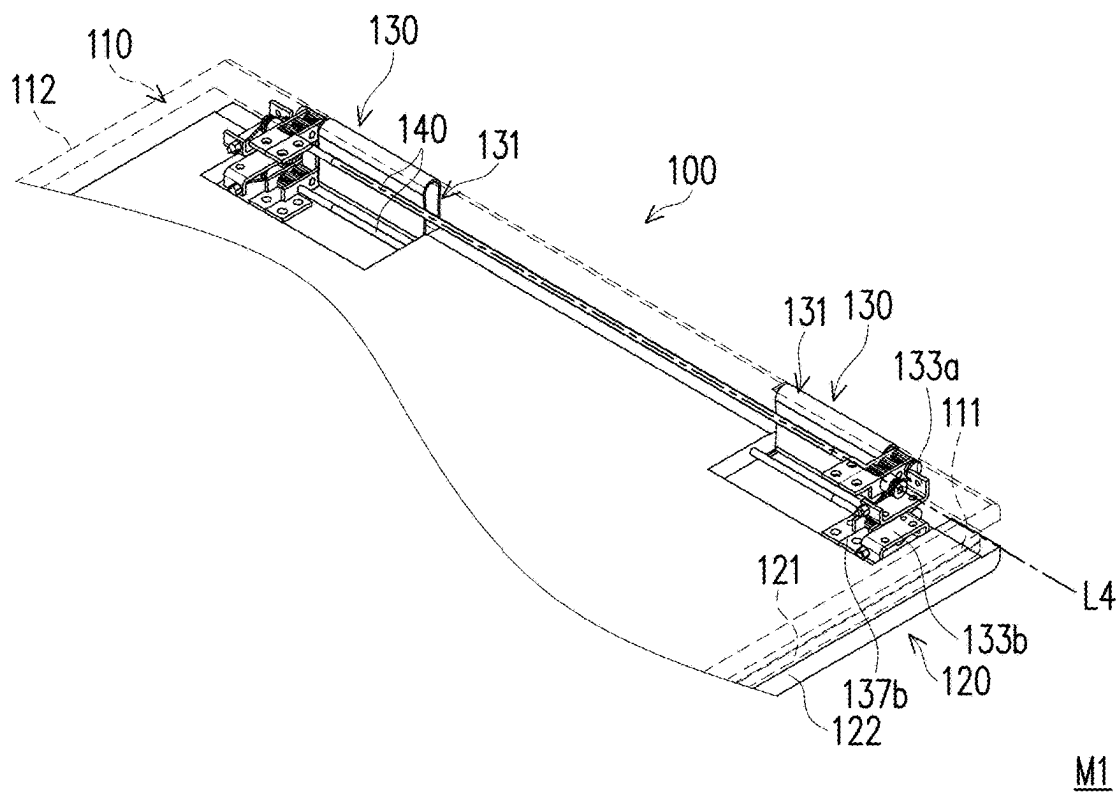


FIG. 2

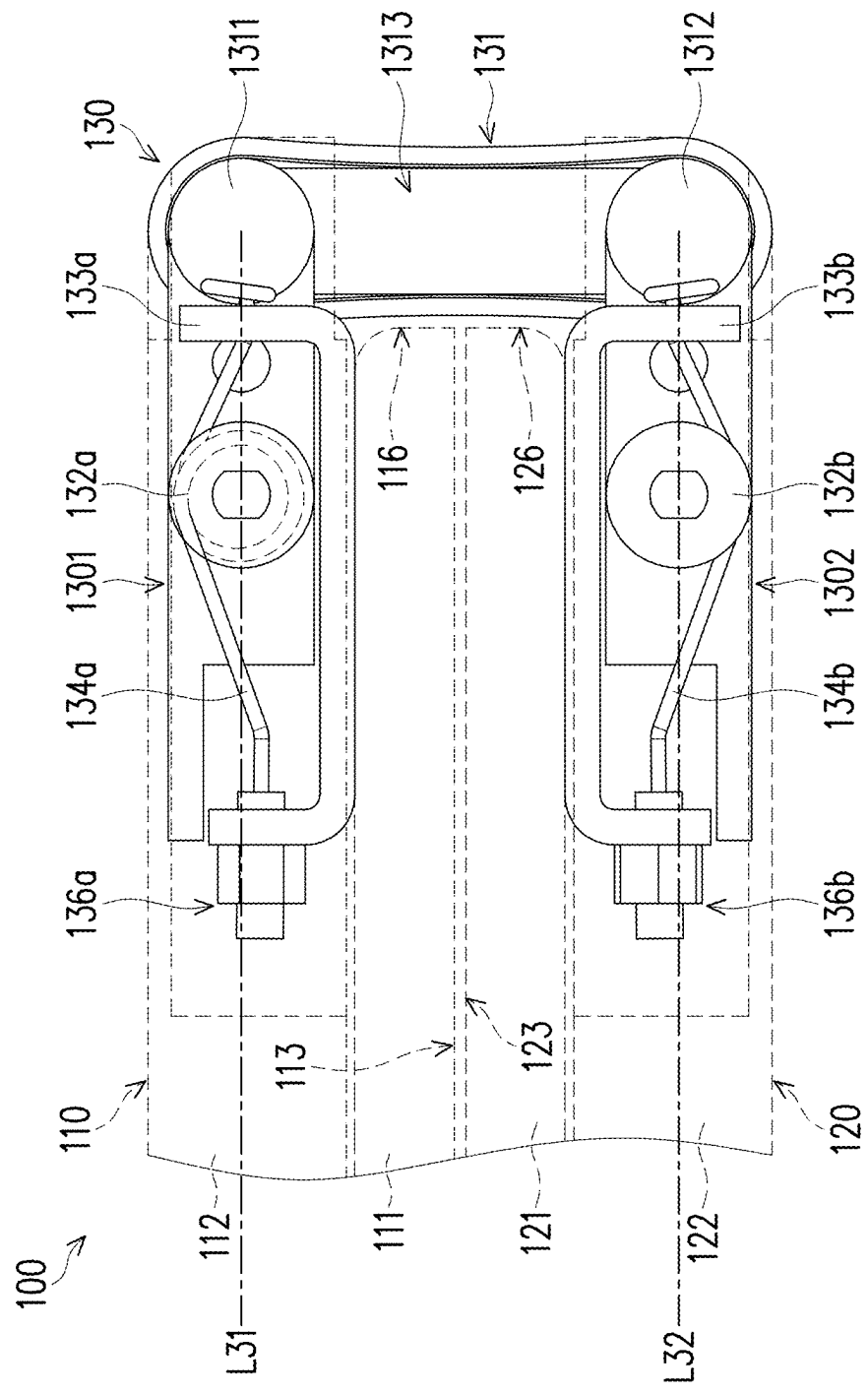


FIG. 3

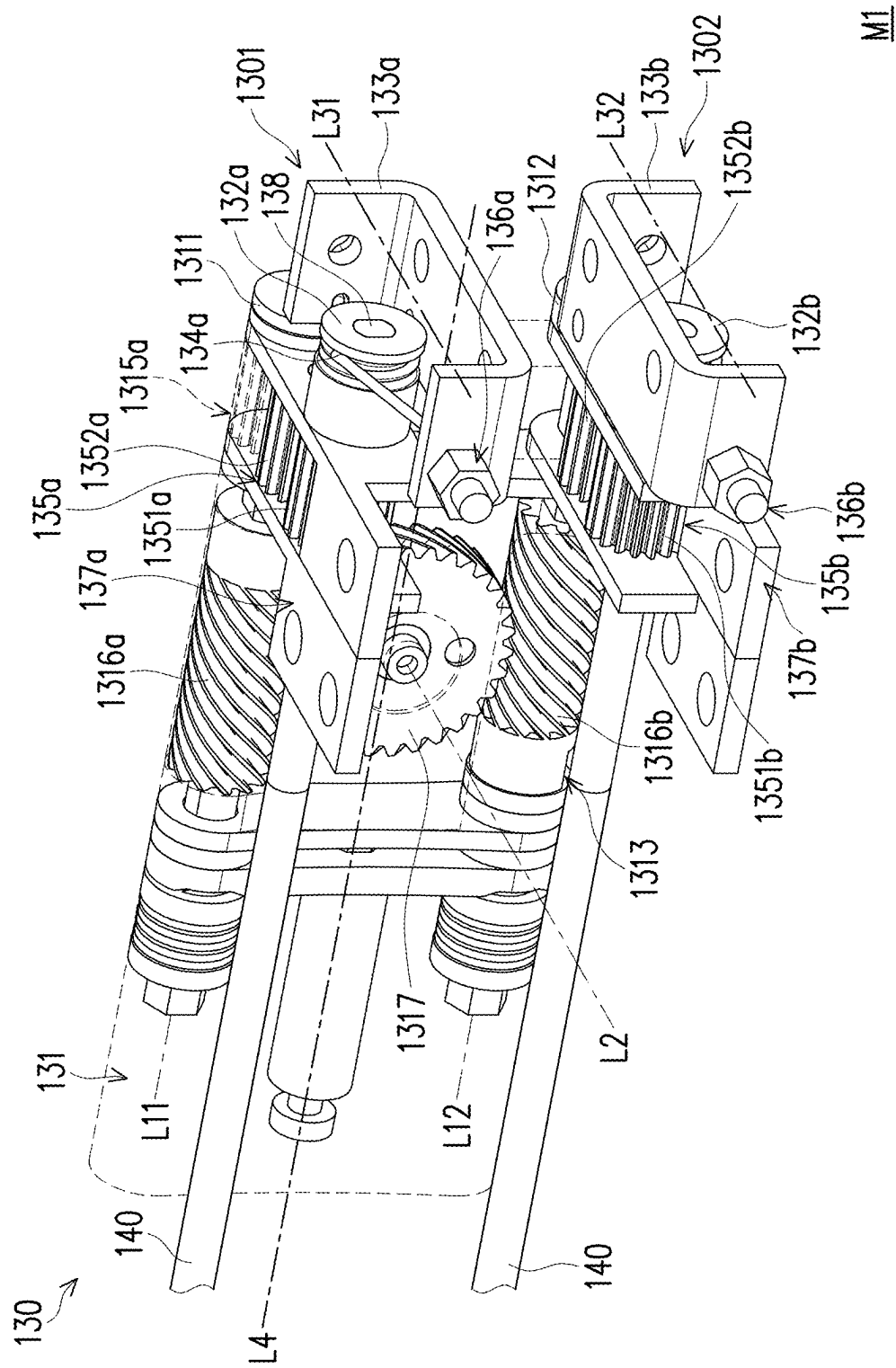


FIG. 4

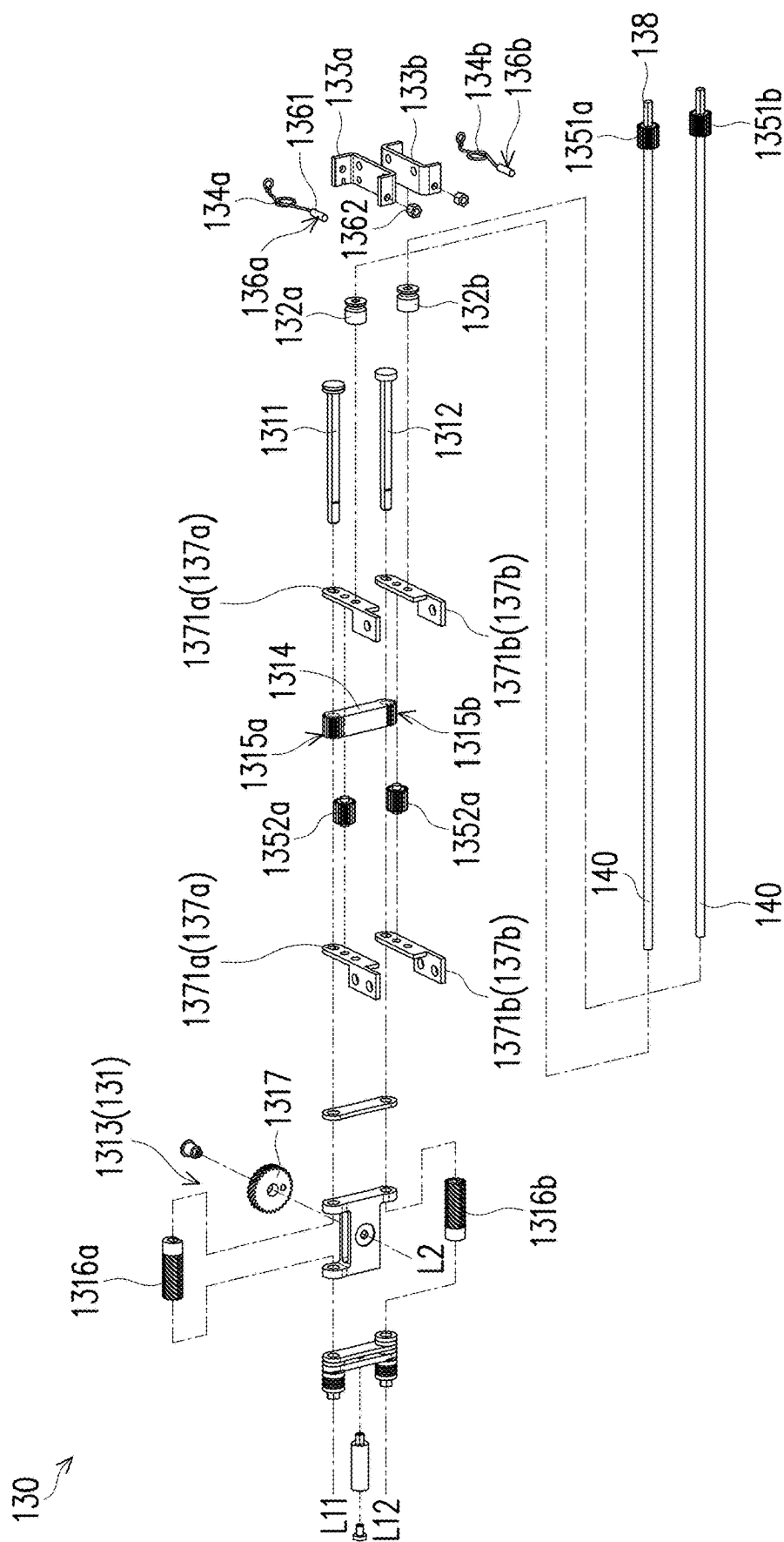


FIG. 5

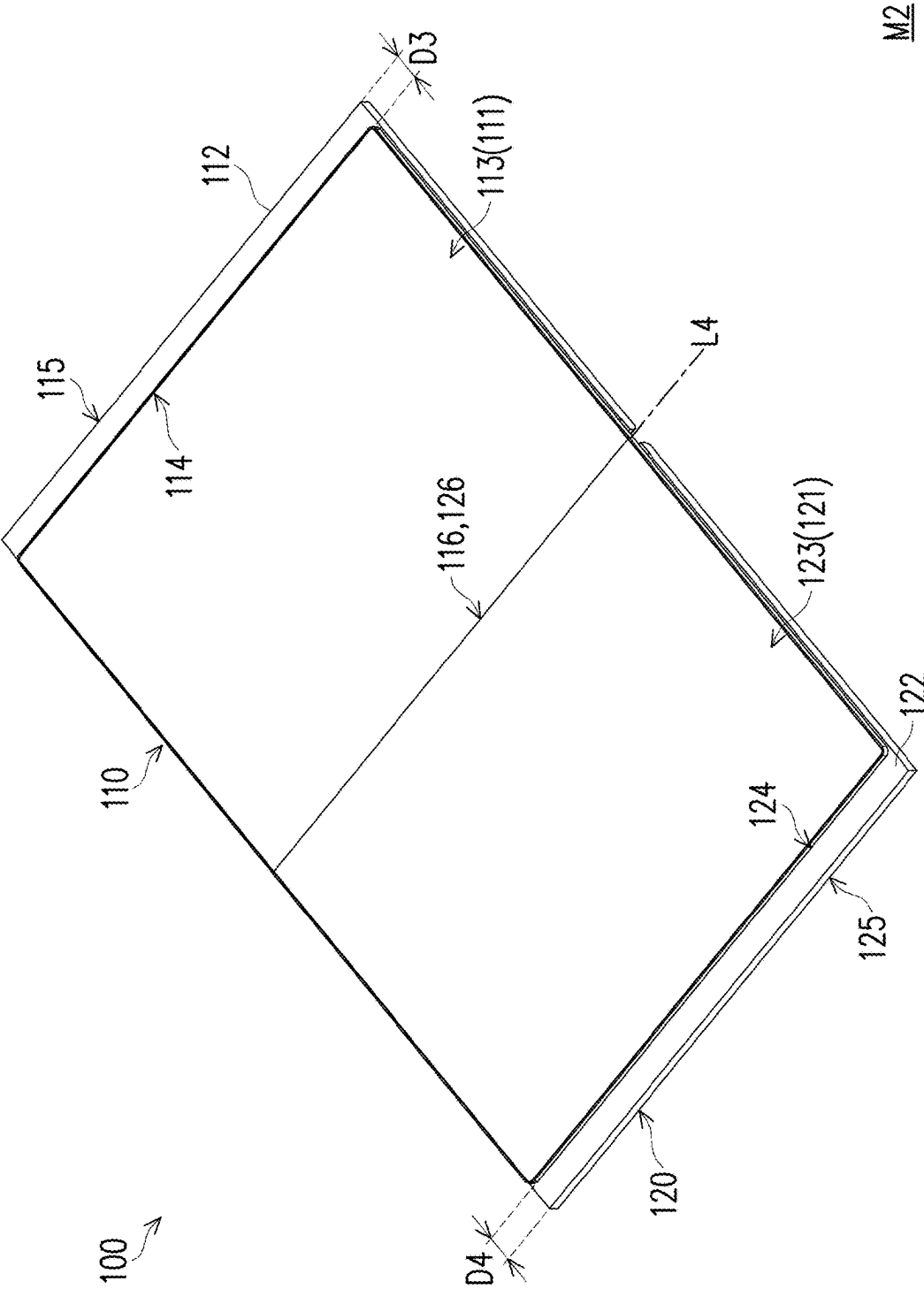


FIG. 6

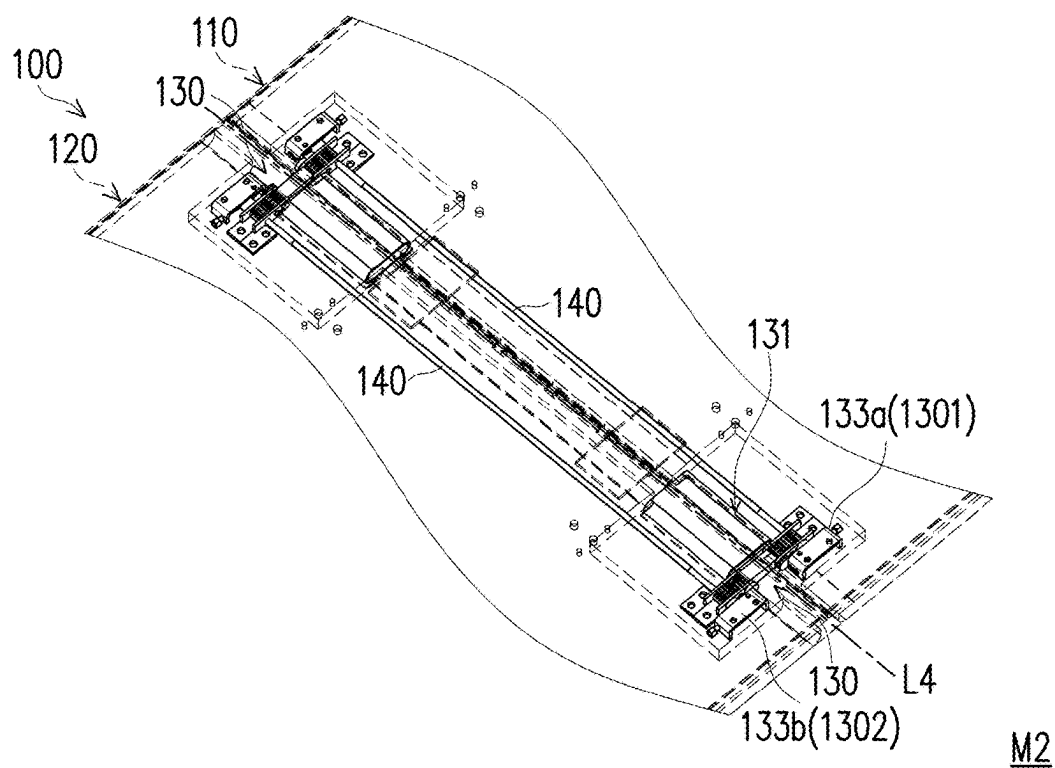


FIG. 7

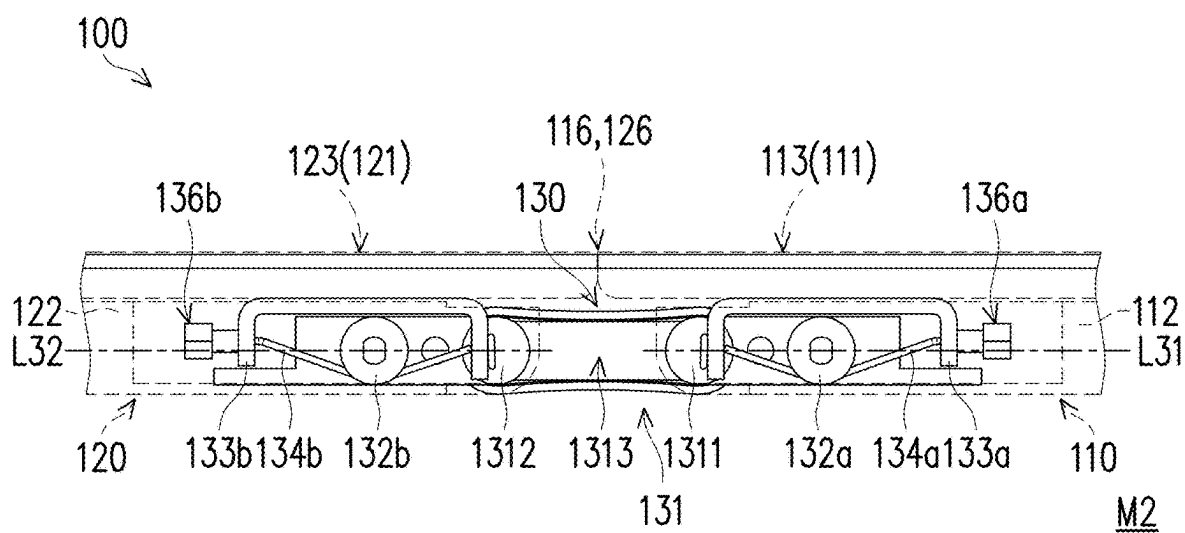
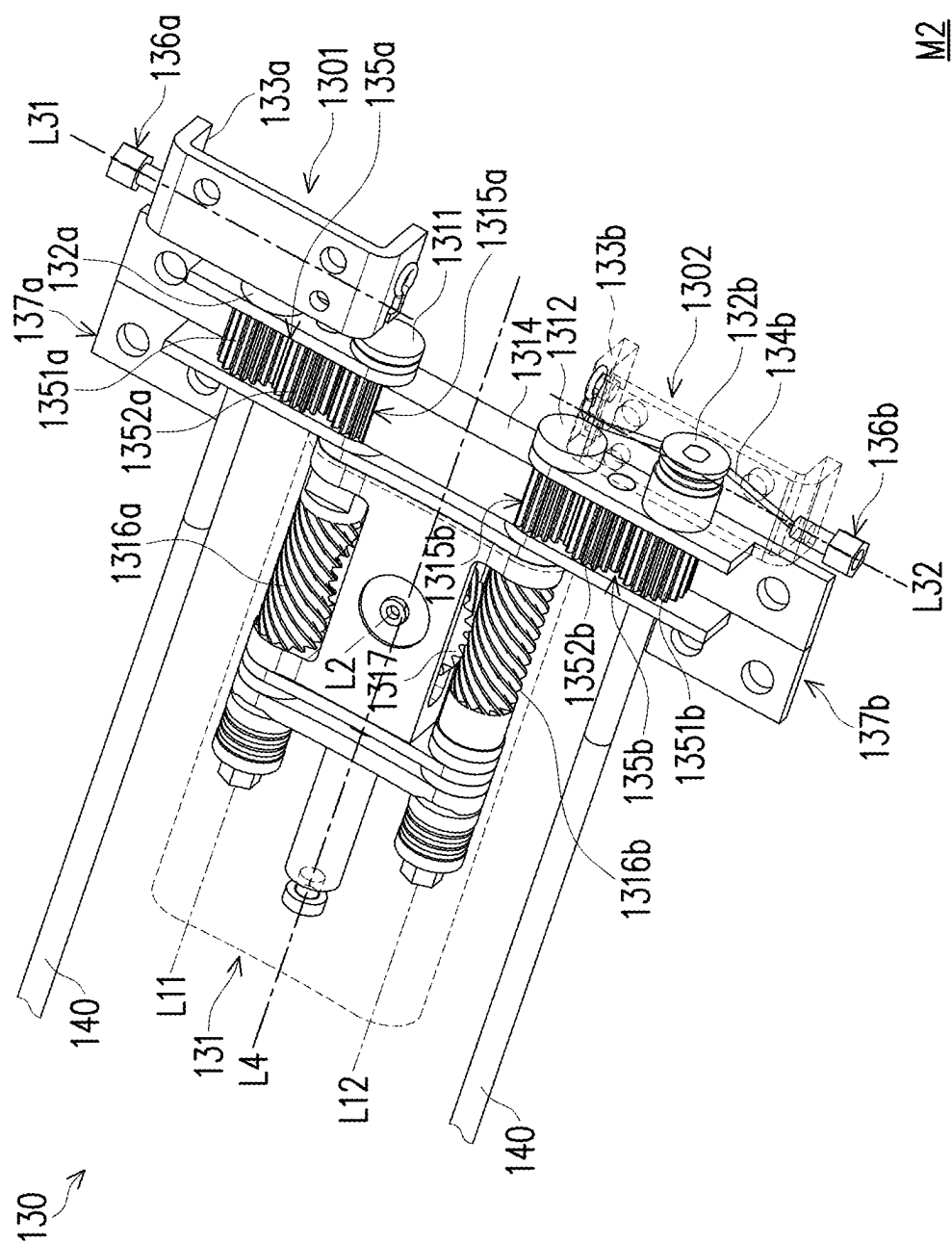


FIG. 8

M2

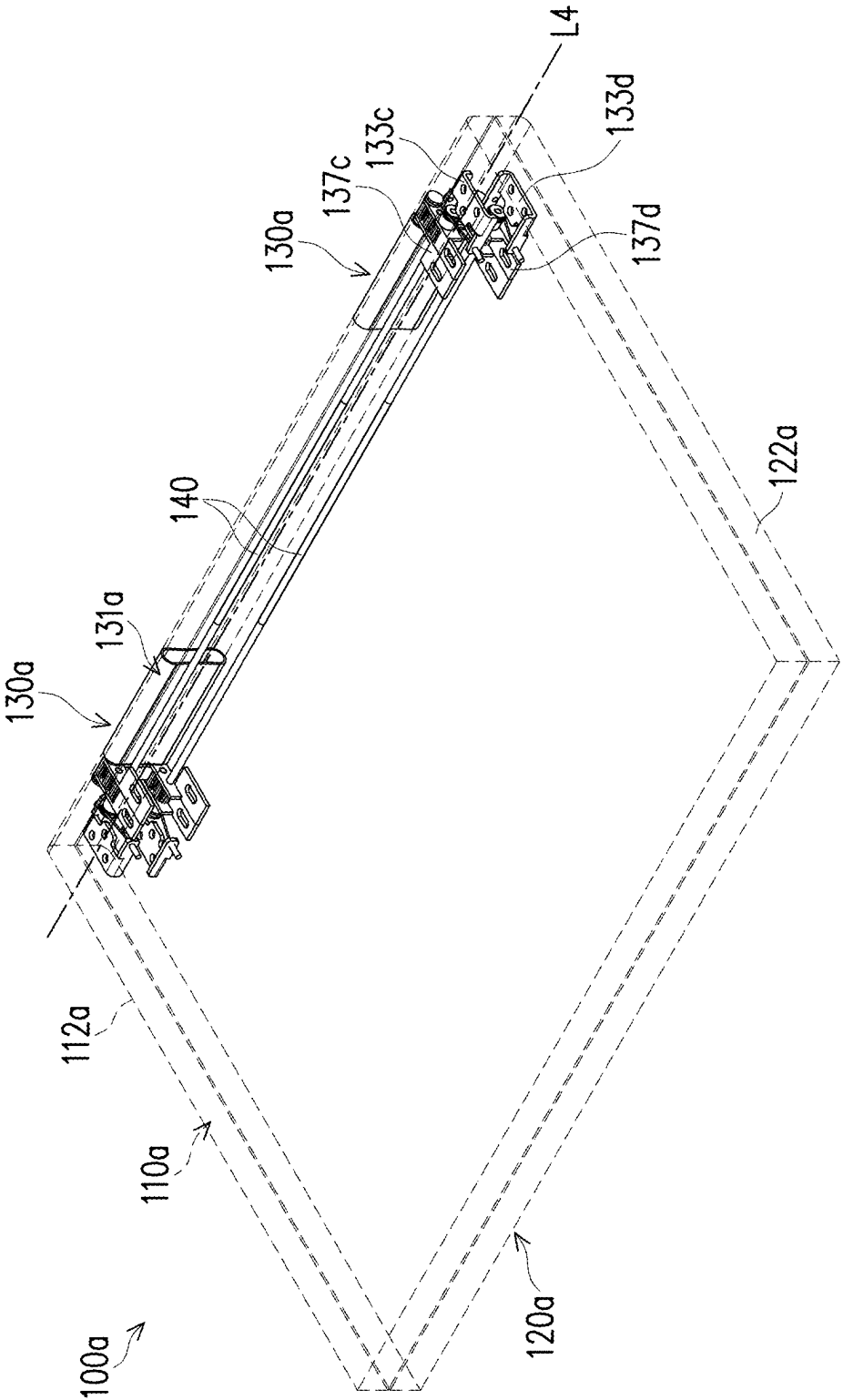


FIG. 10A

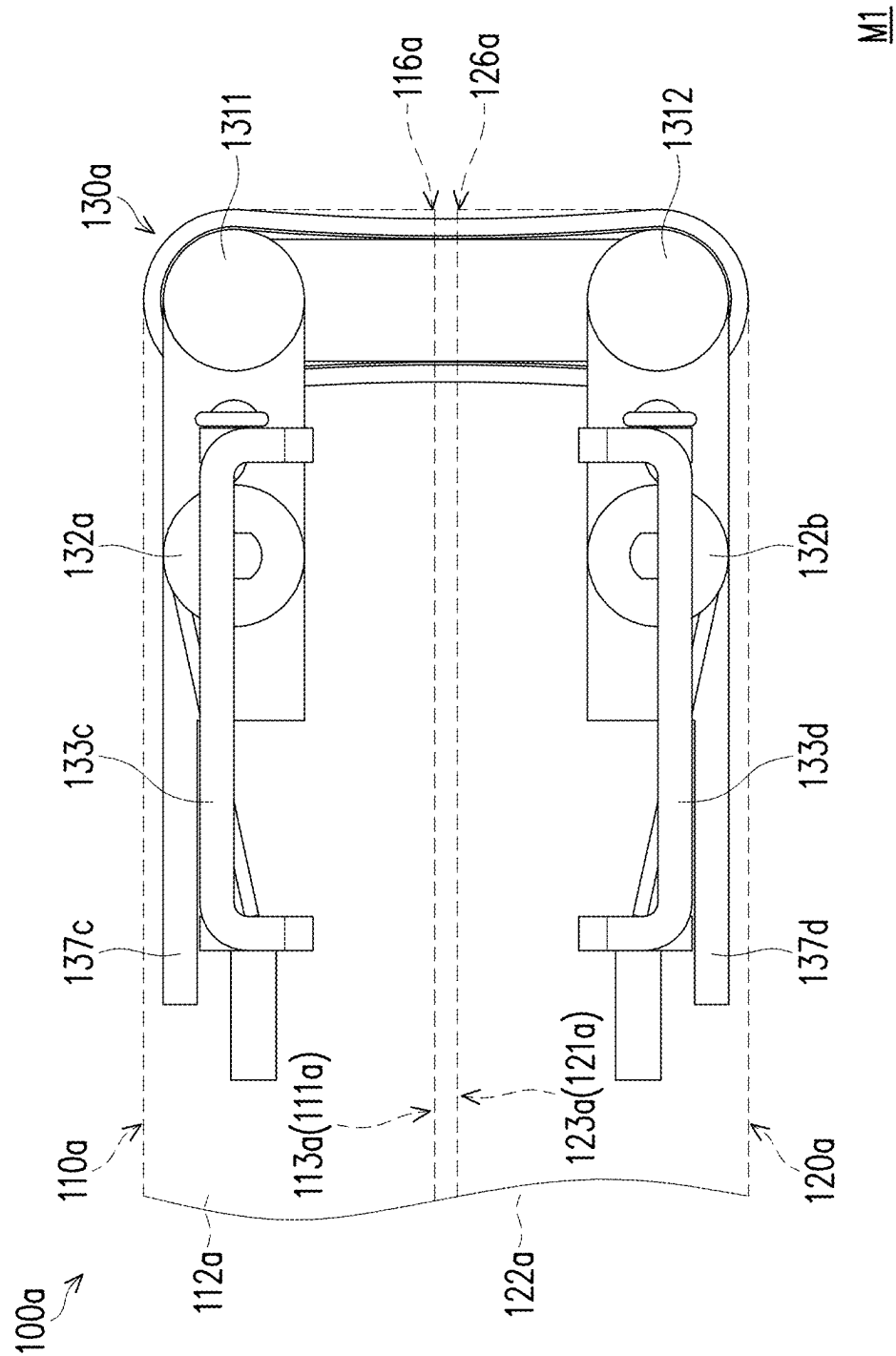


FIG. 10B

FIG. 11

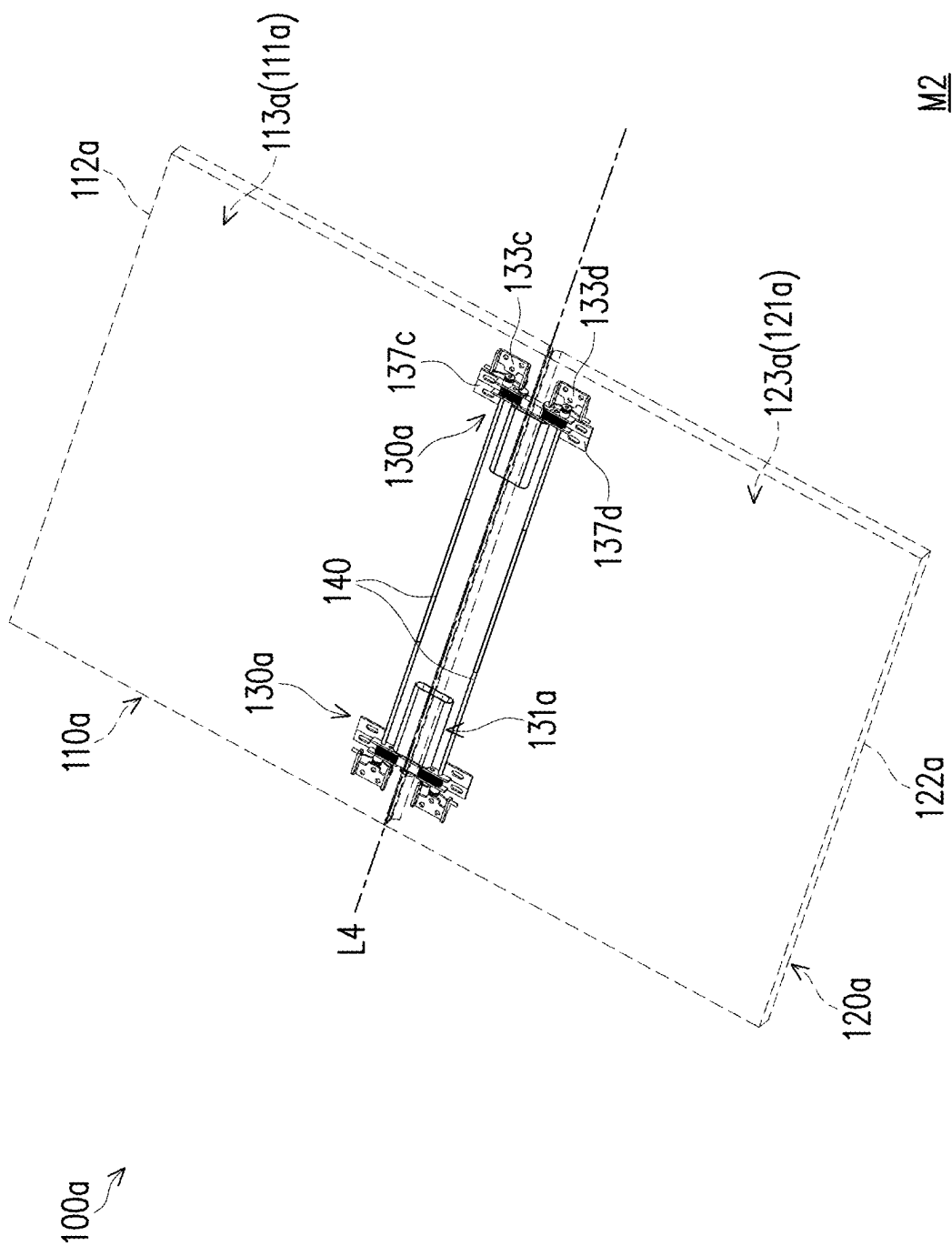


FIG. 12A

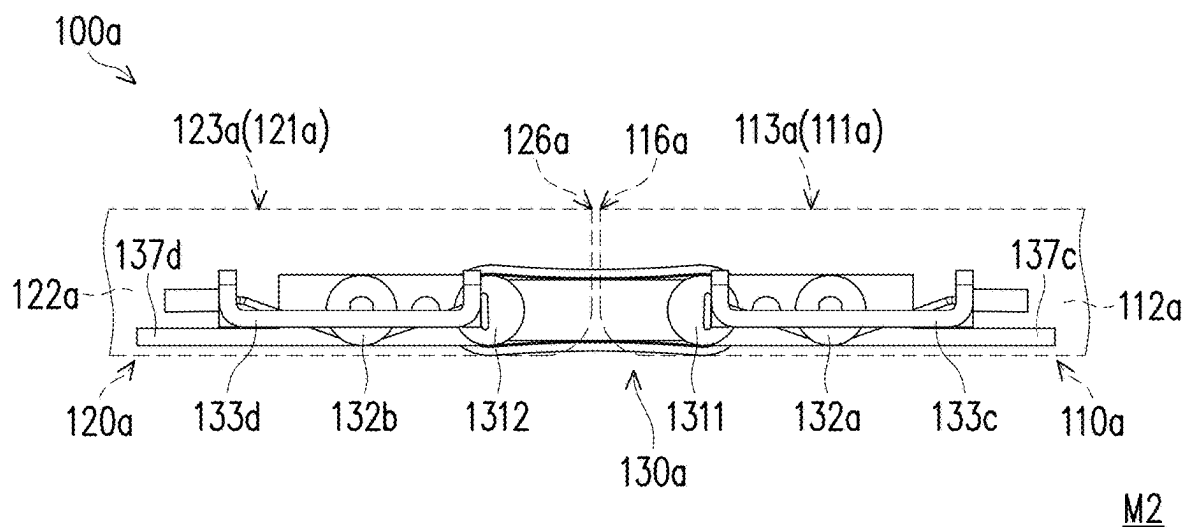


FIG. 12B

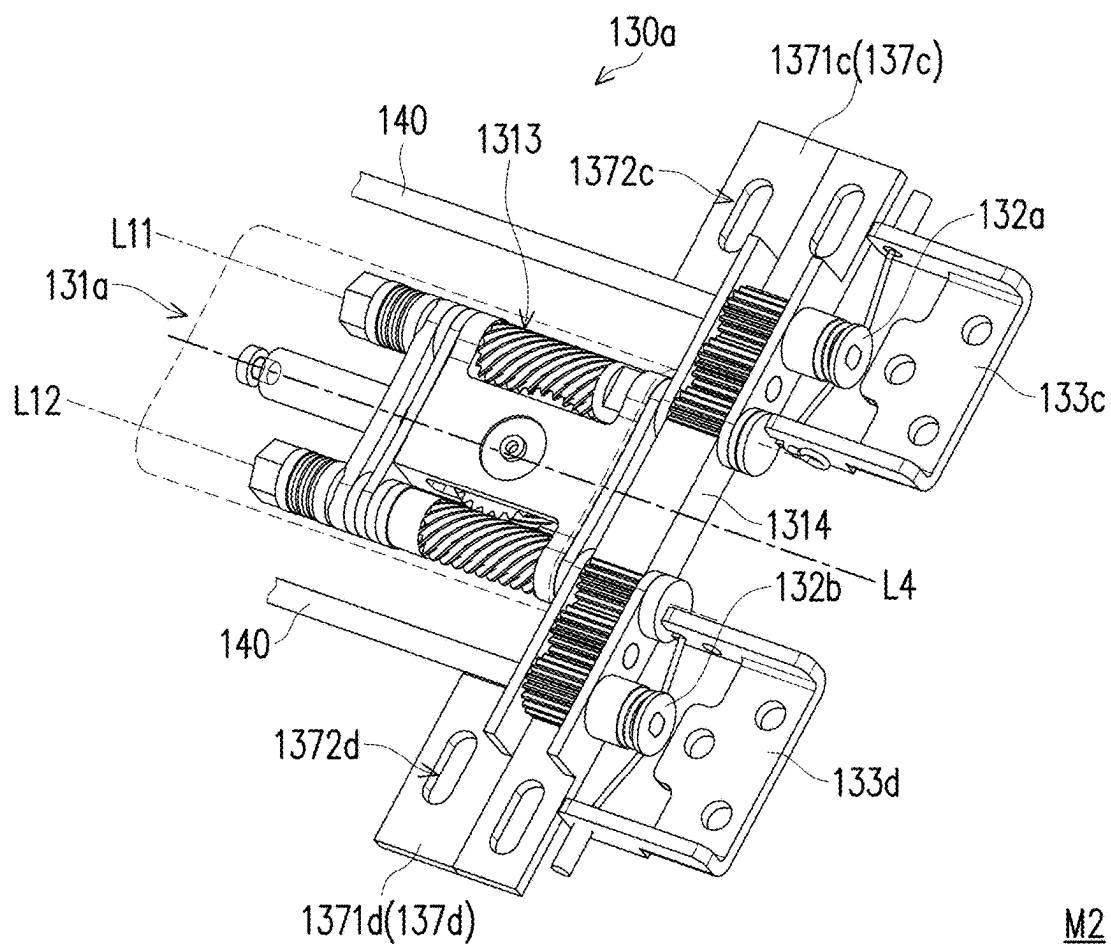


FIG. 13

ELECTRONIC DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of U.S. provisional application Ser. No. 63/561,238, filed on Mar. 4, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

[0002] The disclosure relates to an electronic device.

Description of Related Art.

[0003] The modern electronic device includes two bodies and a pivot, with the two bodies rotatably connected through the pivot. When the two bodies are relatively flipped, there is a gap between the two bodies due to the structural limitations of the pivot, that is unfavorable for users to view and use the electronic device.

SUMMARY

[0004] The disclosure provides an electronic device that is convenient to use.

[0005] The electronic device of the disclosure includes a first body, a second body and at least one pivot module. The first body is pivotably connected to the second body. The pivot module includes a first transmission wheel, a linkage assembly, a first sliding member, a first driven member and a first rope. The linkage assembly is connected to the first body and the second body. The first sliding member is connected to the first body and linked to the first transmission wheel. The first driven member is connected to the first transmission wheel and the linkage assembly.

[0006] The first rope is connected to the first sliding member and the first transmission wheel. The rotation of the first body relative to the second body drives the first driven member and the linkage assembly to move, thereby driving the first transmission wheel to rotate. The rotation of the first transmission wheel relative to the first rope drives the first sliding member and at least part of the first body to slide relative to a central axis of the linkage assembly.

[0007] Based on the above, the first body of the electronic device of the disclosure is connected to the linkage assembly of the pivot module. When the first body rotates, the first body leads the first transmission wheel to rotate through the first driven member and the linkage assembly, and the first transmission wheel drives the first sliding member to slide through the first rope. Thereby, at least part of the first body connected to the first sliding member slides relative to the center axis of the linkage assembly while the first body rotates relative to the second body, to improve the convenience of using the electronic device.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram of an electronic device according to an embodiment of the disclosure.

[0009] FIG. 2 is another schematic diagram of the electronic device in FIG. 1.

[0010] FIG. 3 is a side view of the electronic device in FIG. 1.

[0011] FIG. 4 is a schematic diagram of the pivot module in FIG. 2.

[0012] FIG. 5 is an exploded view of the pivot module in FIG. 4.

[0013] FIG. 6 is a schematic diagram of the electronic device in FIG. 1 in the second status.

[0014] FIG. 7 is a partially enlarged view of the electronic device in FIG. 6.

[0015] FIG. 8 is a side view of the electronic device in FIG. 6.

[0016] FIG. 9 is a schematic diagram of the pivot module in FIG. 7.

[0017] FIG. 10A is a schematic diagram of an electronic device in the first status according to another embodiment of the disclosure.

[0018] FIG. 10B is a side view of the electronic device in FIG. 10A.

[0019] FIG. 11 is a schematic diagram of the pivot module in FIG. 10A.

[0020] FIG. 12A is a schematic diagram of the electronic device in FIG. 10A in the second status.

[0021] FIG. 12B is a side view of the electronic device in FIG. 12A.

[0022] FIG. 13 is a schematic diagram of the pivot module in FIG. 12A.

DESCRIPTION OF THE EMBODIMENTS

[0023] FIG. 1 is a schematic diagram of an electronic device according to an embodiment of the disclosure. FIG. 2 is another schematic diagram of the electronic device in FIG. 1. FIG. 3 is a side view of the electronic device in FIG. 1. FIG. 4 is a schematic diagram of the pivot module in FIG. 2. FIG. 5 is an exploded view of the pivot module in FIG. 4. Please refer to FIG. 1 to FIG. 5 at the same time, the electronic device 100 includes a first body 110, a second body 120, and at least one pivot module 130. The first body 110 is pivotally connected to the second body 120 through the pivot module 130. Each pivot module 130 includes a first driving assembly 1301 and a linkage assembly 131. The linkage assembly 131 is connected to the first body 110 and the second body 120. The first driving assembly 1301 of the pivot module 130 includes a first transmission wheel 132a, a first sliding member 133a, a first driven member 135a, and a first rope 134a. The first driven member 135a is connected to the first transmission wheel 132a and the linkage assembly 131. The first rope 134a is connected to the first sliding member 133a and the first transmission wheel 132a. The first sliding member 133a is connected to the first body 110, and is linked to the first transmission wheel 132a through the first rope 134a.

[0024] The rotation of the first body 110 relative to the second body 120 leads the first driven member 135a and the linkage assembly 131 to move, thereby driving the first transmission wheel 132a to rotate. The rotation of the first transmission wheel 132a relative to the first rope 134a drives the first sliding member 133a and at least part of the first body 110 to slide relative to the second body 120. Thereby, a first function module 111 of the first body 110 may slide relative to a center axis L4 of the linkage assembly 131 and a second function module 121 of the second body 120 through the first driving assembly 1301 while rotating. The electronic device 100 may switch between a first status

M1 (shown in FIG. 1) and a second status M2 (shown in FIG. 6) to improve the convenience of use of the electronic device 100. The number of pivot modules 130 in this embodiment is two, but not limited thereto. The electronic device 100 includes a linkage rod 140, and the two pivot modules 130 are connected by the linkage rod 140, so that the two pivot modules 130 may act synchronously and stably. The first function module 111 and the second function module 121 may include an input module (for example, a keyboard) and/or a display module, but not limited thereto.

[0025] As shown in FIG. 1 to FIG. 3, the first body 110 of this embodiment further includes a first casing 112, and the first function module 111 is slidably disposed on the first casing 112, but not limited thereto. The first sliding member 133a is connected to the first function module 111, and may lead the first function module 111 to slide relative to the first casing 112 along a sliding axis L31. The second body 120 of this embodiment includes a second casing 122, and the second function module 121 of the second body 120 is slidably disposed on the second casing 122, but not limited thereto.

[0026] As shown in FIG. 3 to FIG. 5, the linkage assembly 131 includes a first shaft 1311, a second shaft 1312, a linkage portion 1313, a connecting member 1314, and a first rack structure 1315a. The first shaft 1311 is linked to the second shaft 1312 through the linkage portion 1313. The first shaft 1311 is linked to the first casing 112 of the first body 110, and the second shaft 1312 is linked to the second casing 122 of the second body 120. The first shaft 1311 and the second shaft 1312 are movably sleeved on opposite ends of the connecting member 1314 respectively. The first rack structure 1315a is formed on the end of the connecting member 1314 adjacent to the first shaft 1311. The first shaft 1311 and the second shaft 1312 may rotate relative to the connecting member 1314. The first driven member 135a is connected to the first shaft 1311 and the first transmission wheel 132a, so that the first transmission wheel 132a is linked to the first shaft 1311.

[0027] The linkage portion 1313 of this embodiment includes two main shaft gears 1316a, 1316b and a linkage gear 1317. The two main shaft gears 1316a, 1316b are sleeved on the first shaft 1311 and the second shaft 1312 along the corresponding two main axes L11, L12 respectively. The linkage gear 1317 is disposed between the two main shaft gears 1316a, 1316b along a connecting axis L2 and engages with the two main shaft gears 1316a, 1316b. The two main axes L11, L12 are parallel to each other, and the connecting axis L2 and the sliding axis L31 are perpendicular to the two main axes L11, L12. The center axis L4 of the linkage assembly 131 is intersected with and is perpendicular to the connecting axis L2.

[0028] The linkage assembly 131 further includes a first bracket 137a and a second bracket 137b. The first bracket 137a is connected to the first casing 112 of the first body 110 and includes two components 1371a. The second bracket 137b is connected to the second casing 122 of the second body 120 and includes two components 1371b. The first shaft 1311 is fixedly connected to the first bracket 137a, and may move with the first bracket 137a. The first driven member 135a and the connecting member 1314 (the first rack structure 1315a) are rotatably connected to the first bracket 137a, and are located between the two components

1371a. The second shaft 1312 is fixedly connected to the second bracket 137b, and may move with the second shaft 1312.

[0029] When the first body 110 (the first bracket 137a) rotates relative to the second body 120, the first shaft 1311 is driven to rotate the main shaft gear 1316a, thereby driving the linkage gear 1317 to rotate, which in turn drives the main shaft gear 1316b to rotate, and leading to the rotation of the second shaft 1312 (the second bracket 137b). The second body 120 is driven by the linkage assembly 131 to rotate relative to the first body 110. The first shaft 1311 and the second shaft 1312 may rotate in opposite directions from each other. For example, when the first shaft 1311 is rotated in a clockwise direction, the second shaft 1312 is rotated in a counterclockwise direction. The linkage way of the first shaft 1311 and the second shaft 1312 (i.e., the structure of the linkage assembly 131) is not limited to this embodiment.

[0030] As shown in FIG. 4 and FIG. 5, the first driven member 135a includes a plurality of gears. In this embodiment, the number of gears is two, but not limited thereto. The two gears 1351a, 1352a are engaged with each other. A rod 138 of the gear 1351a is connected to the first transmission wheel 132a, and the gear 1352a is engaged with the first rack structure 1315a of the linkage assembly 131. In this embodiment, the number of teeth of the two gears 1351a, 1352a is the same, but not limited thereto. In an embodiment not shown, the number of teeth of the two gears 1351a, 1352a may be different, to adjust the moving distance of the first sliding member 133a on the sliding axis L31 by the ratio of the number of teeth of the two gears 1351a, 1352a. The linkage rod 140 may be connected to the gear 1351a. In an embodiment not shown, the number of gears of the first driven member 135a may be three, four, or any arbitrary number. In an embodiment not shown, the number of gears of the first driven member 135a may be one, and the gear may be connected to the first transmission wheel 132a and the linkage rod 140, and engaged with the first rack structure 1315a.

[0031] As shown in FIG. 3, the first rope 134a winds around at least part of the first transmission wheel 132a. In this embodiment, the first rope 134a winds around the entire first transmission wheel 132a, in other words, the number of turns of the first rope 134a winding around the first transmission wheel 132a is one turn, but not limited thereto. By adjusting the number of turns of the first rope 134a, the contact area between the first rope 134a and the first transmission wheel 132a may be adjusted, thereby adjusting the friction between the first rope 134a and the first transmission wheel 132a. The first rope 134a starts winding around the first transmission wheel 132a from the edge farther from the first function module 111, and after winding one turn, the first rope 134a leaves the first transmission wheel 132a from the same edge of the first transmission wheel 132a, but not limited thereto.

[0032] The winding way and the number of turns of the first rope 134a are not limited to this embodiment. In an embodiment not shown, the first rope 134a may only abut against a part of the first transmission wheel 132a. That is, the number of turns of the first rope 134a may be less than one turn. In an embodiment not shown, the number of turns of the first rope 134a may be greater than one turn. In an embodiment not shown, the first rope 134a may start winding around the first transmission wheel 132a from the edge

closer to the first function module 111, and leave the first transmission wheel 132a from the same edge of the first transmission wheel 132a.

[0033] The first driving assembly 1301 of the pivot module 130 further includes a first adjustment member 136a. The shape of the first sliding member 133a is, for example, U-shaped, but not limited thereto. One end of the first rope 134a is fixed to the first sliding member 133a, and the other end is connected to the first adjustment member 136a. The first adjustment member 136a is used to adjust the friction between the first transmission wheel 132a and the first rope 134a. Even if the friction between the first transmission wheel 132a and the first rope 134a decreases due to long-term use, the first rope 134a may be maintained in a tightened state through the first adjustment member 136a, thereby maintaining the certain friction between the first transmission wheel 132a and the first rope 134a, allowing the first transmission wheel 132a to rotate and displace stably. The first adjustment member 136a includes, for example, a stick 1361 and a nut 1362, but not limited thereto. The stick 1361 is connected to the first rope 134a and the first sliding member 133a. The nut 1362 is movably connected to the stick 1361, and the movement of the nut 1362 relative to the stick 1361 may adjust the relative position between the stick 1361 and the first sliding member 133a.

[0034] As shown in FIG. 3 to FIG. 5, the pivot module 130 of this embodiment further includes a second driving assembly 1302. The second driving assembly 1302 has a structure similar to the first driving assembly 1301. Specifically, the second driving assembly 1302 of the pivot module 130 includes a second transmission wheel 132b, a second sliding member 133b, a second rope 134b, a second driven member 135b, and a second adjustment member 136b. The linkage assembly 131 further includes a second rack structure 1315b, the second rack structure 1315b is formed on the other end of the connecting member 1314 near the second shaft 1312. The second sliding member 133b is connected to the second function module 121 of the second body 120, and may lead the second function module 121 to slide relative to the second casing 122 along the sliding axis L31.

[0035] The second driven member 135b is connected to the second transmission wheel 132b and the second shaft 1312 of the linkage assembly 131, thereby linking the second transmission wheel 132b to the second shaft 1312. The second driven member 135b includes two gears 1351b, 1352b. The gears 1351b, 1352b are engaged with each other. The gear 1351b is connected to the second transmission wheel 132b, and the gear 1352b is engaged with the second rack structure 1315b. The linkage rod 140 is connected to the gear 1351b. The second sliding member 133b is linked to the second transmission wheel 132b. One end of the second rope 134b is fixedly connected to the second sliding member 133b, and the other end is connected to the second adjustment member 136b. The second rope 134b surrounds and connects to the second transmission wheel 132b. The second driven member 135b and the connecting member 1314 (the second rack structure 1315b) are rotatably connected to the second bracket 137b and located between the two components 1371b.

[0036] The rotation of the first body 110 relative to the second body 120 leads the linkage assembly 131 to move, thereby driving the second body 120 to rotate relative to the first body 110, and leading the second driven member 135b to drive the second transmission wheel 132b to rotate. The

rotation of the second transmission wheel 132b relative to the second rope 134b drives the second sliding member 133b and the second function module 121 of the second body 120 to slide along the sliding axis L32 relative to the first body 110. The two sliding axes L31, L32 are parallel to each other.

[0037] FIG. 1 to FIG. 4 shown the electronic device 100 in the first status M1. In the first status M1, a first outer surface 113 of the first function module 111 of the first body 110 faces towards a second outer surface 123 of the second function module 121 of the second body 120, the first body 110 is parallel to the second body 120, and the electronic device 100 is closed. A projection of the first function module 111 to the first casing 112 is completely located on the first casing 112, and a projection of the linkage assembly 131 to the first casing 112 is completely located on the first casing 112. There is a distance D1 between an end 114 of the first function module 111 and an end 115 of the first casing 112. In this embodiment, the distance D1 is greater than zero, but not limited thereto.

[0038] A projection of the second function module 121 to the second casing 122 is completely located on the second casing 122, and a projection of the linkage assembly 131 to the second casing 122 is completely located on the second casing 122. There is a distance D2 between an end 124 of the second function module 121 and an end 125 of the second casing 122. The distance D1 is equal to the distance D2, but not limited thereto. The first sliding member 133a and the second sliding member 133b are in the position shown in FIG. 4. The sliding axes L31, L32 are parallel to the connecting axis L2.

[0039] FIG. 6 is a schematic diagram of the electronic device in FIG. 1 in the second status. FIG. 7 is a partially enlarged view of the electronic device in FIG. 6. FIG. 8 is a side view of the electronic device in FIG. 6. FIG. 9 is a schematic diagram of the pivot module in FIG. 7. Please refer to FIG. 6 to FIG. 9 at the same time. When switching the electronic device 100 from the first status M1 to the second status M2, the first body 110 (the first bracket 137a) may be forced to rotate relative to the second body 120, thereby leading the first shaft 1311 to rotate. The first driving assembly 1301 (i.e., the first transmission wheel 132a, the first sliding member 133a, the first rope 134a, the first driven member 135a, and the first adjustment member 136a) is led by the first bracket 137a to rotate relative to the first rack structure 1315a of the linkage assembly 131 with the main axis L11 as the axis.

[0040] Since the gear 1352a of the first driven member 135a is engaged with the first rack structure 1315a, the rotation of the first driving assembly 1301 causes the gear 1352a to rotate relative to the first rack structure 1315a and the first bracket 137a, thereby driving the gear 1351a to rotate. The first transmission wheel 132a is led by the gear 1351a to rotate relative to the first rope 134a, to drive the first sliding member 133a to lead the first function module 111 of the first body 110 to slide relative to the first casing 112 along the sliding axis L31.

[0041] During the rotation process of the first shaft 1311, the second shaft 1312 is led by the first shaft 1311 through the linkage portion 1313, causing the second body 120 (the second bracket 137b) to rotate relative to the first body 110. The second driving assembly 1302 (i.e., the second transmission wheel 132b, the second sliding member 133b, the second rope 134b, the second driven member 135b, and the

second adjustment member **136b**) is led by the second bracket **137b** to rotate relative to the second rack structure **1315b**. Thereby, the gears **1351b**, **1352b** of the second driven member **135b** are led by the second rack structure **1315b** to drive the second transmission wheel **132b** to rotate relative to the second rope **134b**, causing the second sliding member **133b** to lead the second function module **121** of the second body **120** to slide relative to the second casing **122** along the sliding axis **L32**. The first function module **111** and the second function module **121** move closer to each other.

[0042] The first body **110** continues to be forced until the first body **110** and the second body **120** relatively move to the position shown in FIG. 6. The electronic device **100** switches to the second status M2. In the second status M2, the angle between the first function module **111** and the second function module **121** is 180 degrees, but not limited thereto. An edge **116** of the first outer surface **113** of the first function module **111** is closely attached to an edge **126** of the second outer surface **123** of the second function module **121**, to increase the continuity of the first outer surface **113** and the second outer surface **123**. The projection of the first function module **111** to the first casing **112** is partially located on the first casing **112**, and the projection of the second function module **121** to the second casing **122** is partially located within the second casing **122**. The first function module **111** and the second function module **121** jointly cover the pivot module **130** from the viewing angle in FIG. 6. The distance D3 between the end **114** of the first function module **111** and the end **115** of the first casing **112** is greater than the distance D1 (shown in FIG. 1) in the first status M1. The distance D4 between the end **124** of the second function module **121** and the end **125** of the second casing **122** is greater than the distance D2 (shown in FIG. 1) in the first status M1. The distance D3 is equal to the distance D4, but not limited thereto. The two sliding axes **L31**, **L32** are perpendicular to the connecting axis **L2**.

[0043] When switching the electronic device **100** from the second status M2 to the first status M1, the first body **110** is forced to rotate towards the second body **120**. The first body **110** leads the first driving assembly **1301** and the linkage assembly **131** to move, causing the first function module **111** and the first casing **112** to rotate, and the first function module **111** to slide relative to the first casing **112**. Simultaneously, the linkage assembly **131** drives the second driving assembly **1302** to move, causing the second function module **121** and the second casing **122** to rotate, and the second function module **121** to slide relative to the second casing **122**. Thereby, the electronic device **100** moves from the position shown in FIG. 8 (the second status M2) to the position shown in FIG. 3 (the first status M1).

[0044] It can be understood that during the switching process of the electronic device **100** between the first status M1 and the second status M2, the first body **110** (the first function module **111** and the first casing **112**) rotates with the main axis **L11** as the axis, and the first function module **111** slides relative to the first casing **112** along the sliding axis **L31**. The second body **120** (the second function module **121** and the second casing **122**) rotates with the main axis **L12** as the axis, and the second function module **121** slides relative to the second casing **122** along the sliding axis **L32**.

[0045] FIG. 10A is a schematic diagram of an electronic device in the first status according to another embodiment of the disclosure. FIG. 10B is a side view of the electronic device in FIG.

[0046] 10A. FIG. 11 is a schematic diagram of the pivot module in FIG. 10A. FIG. 12A is a schematic diagram of the electronic device in FIG. 10A in a second status. FIG. 12B is a side view of the electronic device in FIG. 12A. FIG. 13 is a schematic diagram of the pivot module in FIG. 12A. Please refer to FIG. 2 and FIG. 10A to FIG. 13 at the same time, the electronic device **100a** of this embodiment is similar to the previous embodiment. The difference between the two is that the first sliding member **133c** of the linkage assembly **131a** of this embodiment is connected to the first casing **112a** of the first body **110a** to lead the first body **110a** to slide relative to the center axis **LA** of the linkage assembly **131a** and the second body **120a**. The second sliding member **133d** is connected to the second casing **122a** of the second body **120a** to lead the second body **120a** to slide relative to the first body **110a**. The first function module **111a** is fixedly disposed on the first casing **112a**, and the second function module **121a** is fixedly disposed on the second casing **122a**. The first bracket **137c** includes a slot **1372c** extending in a direction parallel to the sliding axis **L31**, and the second bracket **137d** includes a slot **1372d** extending in a direction parallel to the sliding axis **L31**. The slots **1372c**, **1372d** are formed on the components **1371c**, **1371d**.

[0047] The edge **116a** of the first outer surface **113a** is an end of the first casing **112a**. The edge **126a** of the second outer surface **123a** is an end of the second casing **122a**. A fastener (not shown) passes through the slot **1372c** to connect the first bracket **137c** to the first casing **122a**, and another fastener (not shown) passes through the slot **1372d** to connect the second bracket **137d** to the second casing **122b**. When the electronic device **100a** switches between the first status M1 and the second status M2, the first body **110a** (the first casing **112a**) rotates relative to the second body **120a** with the main axis **L11** as the axis, and is led by the first sliding member **133c** to slide relative to the first bracket **137c** along the sliding axis **L31**. The second body **120a** (the second casing **122a**) rotates relative to the first body **110a** with the main axis **L12** as the axis, and is led by the second sliding member **133d** to slide relative to the second bracket **137d** along the sliding axis **L32**. The fasteners slide in the slots **1372c**, **1372d**. The electronic device **100a** of this embodiment has similar effect to the previous embodiment, and is not repeated herein.

[0048] The movement way of the first body **110**, **110a** and the second body **120**, **120a** of the electronic device **100**, **100a** is not limited to the above embodiments. For example, in an embodiment not shown, the pivot module **130**, **130a** may only include the linkage assembly **131** and the first driving assembly **1301** (i.e., the first transmission wheel, the first sliding member, the first rope, the first driven member, and the first adjustment member). When the electronic device **100**, **100a** switches between the first status M1 and the second status M2, the first body **110**, **110a** may move relative to the second body **120**, **120a** through the first driving assembly **1301** in any way of the above embodiments, while the second body **120**, **120a** may only be driven by the linkage assembly **131** to rotate relative to the first body **110**, **110a**. That is, only the first outer surface **113**, **113a** may move towards the second outer surface **123**, **123a**, while the second outer surface **123**, **123a** does not move towards the first outer surface **113**, **113a**.

[0049] In summary, the first body of the electronic device of the disclosure is connected to the linkage assembly of the pivot module. When the first body rotates, the first body

leads the first transmission wheel to rotate through the first driven member and the linkage assembly, and the first transmission wheel drives the first sliding member to slide through the first rope. Thereby, at least part of the first body connected to the first sliding member slides relative to the center axis of the linkage assembly while the first body rotates relative to the second body, to improve the convenience of using the electronic device.

What is claimed is:

1. An electronic device, comprising:

a first body;

a second body, the first body is pivotally connected to the second body; and

at least one pivot module, wherein each of the at least one pivot module comprises:

a first transmission wheel;

a linkage assembly, connected to the first body and the second body;

a first sliding member, connected to the first body and linked to the first transmission wheel;

a first driven member, connected to the first transmission wheel and the linkage assembly; and

a first rope, connected to the first sliding member and the first transmission wheel,

the rotation of the first body relative to the second body leads the linkage assembly and the first driven member to move, thereby driving the first transmission wheel to rotate, and the rotation of the first transmission wheel relative to the first rope drives the first sliding member and at least a part of the first body to move relative to a center axis of the linkage assembly.

2. The electronic device according to claim 1, wherein the first body comprises a first function module and a first casing, the first function module is slidably connected to the first casing, the first sliding member is connected to the first function module, and is adapted to lead the first function module to slide relative to the first casing.

3. The electronic device according to claim 2, wherein when the electronic device is in a first status, a projection of the first function module to the first casing is completely located on the first casing, and when the electronic device is in a second status, a projection of the first function module to the first casing is partially located on the first casing.

4. The electronic device according to claim 2, wherein a distance between an end of the first function module and an end of the first casing when the electronic device is in a first status is smaller than a distance between the end of the first function module and the end of the first casing when the electronic device is in a second status.

5. The electronic device according to claim 1, wherein the first body comprises a first function module and a first casing, the first function module is disposed at the first casing, the first sliding member is connected to the first casing, and is adapted to lead the first body to slide relative to the second body.

6. The electronic device according to claim 1, wherein the linkage assembly comprises a first shaft, a second shaft and a linkage portion, the first shaft is linked to the first body, the second shaft is linked to the second body, the linkage portion comprises two main shaft gears and a linkage gear, the two main shaft gears are respectively sleeved on the first shaft and the second shaft, the linkage gear is disposed between the two main shaft gears and engages with the two main shaft gears.

7. The electronic device according to claim 6, wherein the linkage assembly comprises a first bracket and a second bracket, the first bracket is connected to the first body, the second bracket is connected to the second body, the first shaft is connected to the first bracket, and the second shaft is connected to the second bracket.

8. The electronic device according to claim 6, wherein the first driven member comprises a plurality of gears engaging with each other, the linkage assembly comprises a connecting member and a first rack structure, the first shaft and the adjustment shaft are movably sleeved on the connecting member, the first rack structure is formed on the connecting member, one of the gears is connected to the first transmission wheel, and another of the gears engages with the first rack structure.

9. The electronic device according to claim 1, wherein each of the at least one pivot module further comprises a first adjustment member, the first rope is connected to the first sliding member and the first adjustment member.

10. The electronic device according to claim 1, wherein each of the at least one pivot module further comprises a second transmission wheel, a second sliding member, a second driven member and a second rope, the second driven member is connected to the second transmission wheel and the linkage assembly, the second sliding member is connected to the second body and linked to the second transmission wheel, the second rope is connected to the second sliding member and the second transmission wheel.

11. The electronic device according to claim 10, wherein the rotation of the first body relative to the second body leads the linkage assembly to move, thereby driving the second body to rotate relative to the first body, and leads the second driven member to drive the second transmission wheel to rotate, the rotation of the second transmission wheel relative to the second rope drives the second sliding member and at least part of the second body to slide relative to the first body.

12. The electronic device according to claim 10, wherein the second body comprises a second function module and a second casing, the second function module is slidably connected to the second casing, the second sliding member is connected to the second function module, and is adapted to lead the second function module to slide relative to the second casing.

13. The electronic device according to claim 12, wherein when the electronic device is in a first status, a projection of the second function module to the second casing is completely located on the second casing, when the electronic device is in a second status, a projection of the second function module to the second casing is partially located on the second casing.

14. The electronic device according to claim 12, wherein a distance between an end of the second function module and an end of the second casing when the electronic device is in a first status is smaller than a distance between the end of the second function module and the end of the second casing when the electronic device is in a second status.

15. The electronic device according to claim 10, wherein the second body comprises a second function module and a second casing, the second function module is disposed at the second casing, the second sliding member is connected to the second casing, and is adapted to lead the second body to slide relative to the first body.

16. The electronic device according to claim **10**, wherein the linkage assembly comprises a first shaft, a second shaft, a linkage portion, a connecting member and a second rack structure, the first shaft is linked to the second shaft through the linkage portion, the second driven member comprises a plurality of gears engaging with each other, the first shaft and the second shaft are movably sleeved on the connecting member, the second rack structure is formed on the connecting member, one of the gears is connected to the second transmission wheel, another of the gears engages with the second rack structure.

17. The electronic device according to claim **10**, wherein each of the at least one pivot module further comprises a second adjustment member, the second rope is connected to the second sliding member and the second adjustment member.

18. The electronic device according to claim **1**, further comprising a linkage rod, the at least one pivot module comprises two pivot modules, the linkage rod is connected to the two pivot modules.

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